(ii).To reconstruct periodic signals using Fourier series and verify the reconstructed waveform with the original signal.

Software / Laboratory Environment:

MATLAB

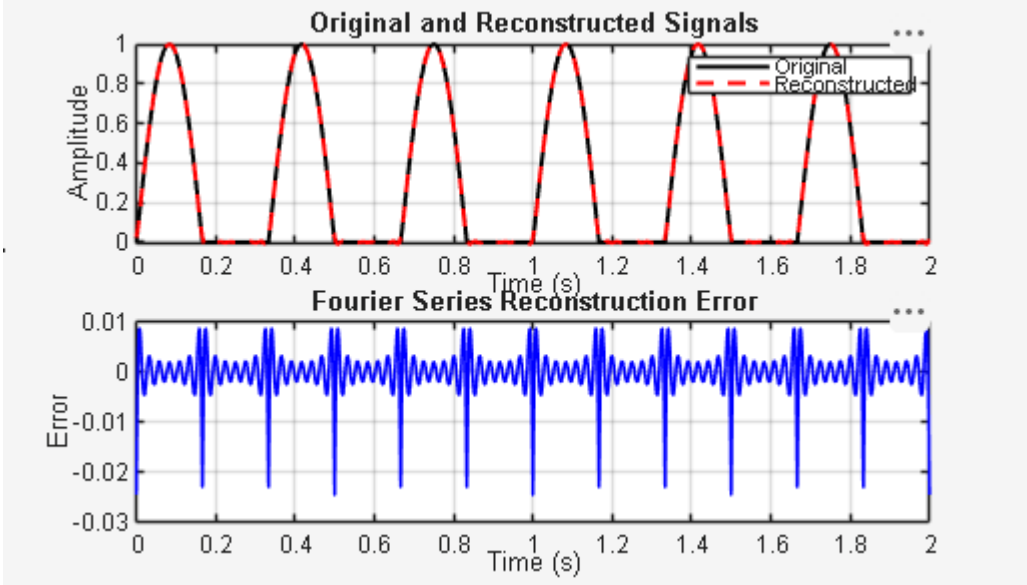
MATLAB Coding:

```
clc;
clear;
close all;


f0 = 3;
T = 3/f0;
t = linspace(0, 2*T, 4001);
x = max(sin(2*pi*f0*t), 0);
N = 12;
tt = t(1:2001);
xx = x(1:2001);
a0 = (2/T)*trapz(tt, xx);
an = zeros(1,N);
bn = zeros(1,N);
for k = 1:N
    an(k) = (2/T)*trapz(tt, xx.*cos(2*pi*k*f0*tt));
    bn(k) = (2/T)*trapz(tt, xx.*sin(2*pi*k*f0*tt));
end
xr = (a0/2)*ones(size(t));
for k = 1:N
    xr = xr + an(k)*cos(2*pi*k*f0*t) ...
        + bn(k)*sin(2*pi*k*f0*t);
end
err = x - xr;
figure(1);
subplot(2,1,1);
plot(t, x, 'k', 'LineWidth', 1.5);
hold on;
```

```
plot(t, xr, 'r--', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Original and Reconstructed Signals');
legend('Original', 'Reconstructed');
subplot(2,1,2);
plot(t, err, 'b', 'LineWidth', 1.2);
grid on;
xlabel('Time (s)');
ylabel('Error');
title('Fourier Series Reconstruction Error');
fprintf('Number of harmonics = %d\n', N);
fprintf('DC component = %.4f\n', a0/2);
fprintf('Mean squared error = %.6f\n', mean(err.^2));
drawnow;
```

Output Graphs:



Conclusion:

The Fourier series representation of periodic continuous-time signals was successfully analysed and the Fourier coefficients and harmonic components were determined. The amplitude and phase spectra were generated and their characteristics were studied. The periodic signal was reconstructed using

Fourier series and the reconstructed waveform was verified with the original signal.

Rubrics:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
(i) To analyze the Fourier series of continuous-time periodic signals and analyze their waveform characteristics.	Fourier Series Analysis	Correctly determines the fundamental period and frequency and computes the Fourier coefficients for every given periodic signal, analytically or in MATLAB.	Computes the coefficients for most signals correctly, with minor errors in limits, algebra, code or parameters.	Computes the coefficients for only some signals, or has noticeable errors in integration, limits or code, and needs guidance.	Unable to compute the Fourier coefficients of the given signals.	20	2	

		Uses suitable commands with correct limits and obtains accurate coefficient values.						
	Waveform Characteristics & Spectrum Analysis	Correctly identifies symmetry (even, odd, half-wave), DC value and the harmonics present. Plots the amplitude and	Identifies most waveform characteristics and plots the spectra correctly, with minor errors or limited explanation.	Identifies basic characteristics. Spectra have noticeable errors or the explanation is incomplete, and needs guidance.	Unable to identify the waveform characteristics or plot the spectra.	20	2	

		phase spectra with titles, axis labels and grid. Explains how symmetry and the shape of the waveform determine the harmonic content and the rate of decay of the coefficients.						
(ii) To reconstruct periodic signals using the	Signal Reconstruction using Fourier Series	Correctly synthesizes each periodic signal	Reconstructs the signals correctly with minor errors	Reconstructs only some signals or uses a single	Unable to reconstruct the signals using the	20	2	

Fourier series and verify the reconstructed waveform with the original signal.		as a partial sum of N harmonics in MATLAB, using a suitable time vector and loop or vectorized code. Reconstructs with several values of N and plots them clearly against the original.	in code, time range, parameters or labelling, and uses fewer values of N.	value of N, with noticeable errors in code or plots, and needs assistance.	Fourier series.			

	Verification of Reconstructed Waveform	Verifies the reconstruction against the original waveform using overlay plots and numerical error for each N. Shows that the error decreases as N increases, and identifies the Gibbs phenomenon at discontinuities with a	Verifies the reconstruction with overlay plots and some error measure, with minor errors in calculation or interpretation of convergence.	Performs basic visual comparison only, with noticeable errors or no numerical error. Convergence or Gibbs phenomenon is not explained, and needs guidance.	Unable to verify the reconstructed waveform against the original signal.	20	2	

		correct explanation.						
Interpret the Fourier series results and maintain the record.	Engineering Analysis, Record &Interpretation	Provides technically accurate interpretation of the results, relating the number of harmonics, symmetry and discontinuities to reconstruction accuracy. Maintains a complete, well-	Maintains a proper record with mostly accurate interpretation, with minor omissions in observations or conclusions.	Record and interpretation are basic or partly incomplete, with limited technical reasoning.	Incomplete record, limited or incorrect interpretation of the results.	20	2	

		organized record with code, plots, observations and conclusions.						
					Total	100	10	